Venkata Lakshmi Parimala Pasupuleti

HUD Compliance & Technical Analyst · SQL · Data Fixes · HUD · HAP Allocation · Tax Credit · Property Management SaaS · Troubleshooting · PHP · Python · AWS

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PROFESSIONAL SUMMARY

8 years at Entrata — Yardi's primary competitor — owning HUD compliance workflows, SQL data pipelines, and affordable housing modules from inside the stack. Built and maintained **HAP allocation, HUD payment processing, income eligibility verification, and housing subsidy compliance** across 100K+ unit portfolios with zero compliance violations. Processed **1.6M HUD/Tax Credit income limit records** via SQL import scripts; designed data normalization and data validation layers; created **56 state-specific certification documents**; built internal debugging tools reducing resolution time 40%; optimized 200+ SQL queries across all client environments. Deep understanding of the property management SaaS domain, federal housing regulations, and the technical issues clients face. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University

Aug 2024 – Aug 2025

Research Associate — Data Pipelines · Analytics · Technical Documentation

- Built multi-source data pipelines (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; designed analytics dashboards (Tableau, Quarto) driving **40% improvement** in faculty decision comprehension; documented technical specifications and methodology across 3 research pipelines.

Entrata — Real Estate PropTech SaaS (Yardi Competitor)

Sep 2018 – Aug 2024

Senior Software Engineer — HUD Compliance · Affordable Housing · SQL · Data Fixes · Financial Operations · Client Modules

- Owned **HUD compliance and affordable housing workflows end-to-end** across 20+ production modules: **HAP allocation, HUD payment processing, gross rent updates, rent schedule management, housing subsidy compliance, rental subsidies, ledger management, income eligibility verification, AMI threshold calculations, Section 8/HCV recertification, LIHTC/Section 42 tax credit calculations**, and payment processing — **zero compliance violations** across 100K+ unit portfolios over 6+ years of production.
- Processed **1.6M HUD and Tax Credit income limit records** via production AI pipeline (Python, PyTorch, OpenAI API) replacing manual Excel workflows — custom neural network architecture for transformation and validation of complex income limit datasets across family size, income levels, and AMI percentage thresholds; also designed original SQL import scripts for same dataset across 5 years of data.
- Wrote **data fixes to correct data corruption** caused by application issues and third-party integrations; designed data normalization and validation layers across all client environments; **optimized 200+ SQL queries** (30% performance gain) via execution plan tuning, indexing, database partitioning, caching, triggers, log tables, and audit tables; led database architecture decisions for complex production schemas.
- Built **microservices for HUD-compliant financial operations**: rent schedules, gross rent updates, HUD-compliant tax credit calculations — integrated into main product ensuring state and federal compliance; led Docker microservices redesign (50% scalability improvement, 30% deployment reduction); achieved **100% PHI compliance** on AWS S3 migration with Libsodium encryption.
- Created **56 state-specific housing certification documents** with merge fields for Section 8/LIHTC/HUD certifications integrated with AWS S3; migrated thousands of client documents to AWS S3 resolving 90% of legacy space constraints; built **internal debugging tools** reducing debugging time 40%; maintained bug count <10 for 6+ months; **300+ code reviews, 5+ engineers mentored**; 100K+ MAU, 99.5% uptime, 120+ Agile sprints.

Entrata India

Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — HUD Data Import · SQL · Document Management · AWS S3

- Built SQL import scripts for **1.6M HUD/Tax Credit income limit records** across 5 years of data; migrated thousands of client documents to AWS S3 resolving 90% of legacy space constraints; developed full-stack property management modules (PHP, PostgreSQL, JavaScript); participated in sprint planning, code reviews, and quality assurance testing.

KEY HUD & COMPLIANCE PROJECTS

HUD/Tax Credit Income Limits Import & Eligibility Automation

HUD PROPTECH

SQL · PHP · PostgreSQL · PL/SQL · Python · Data Normalization · Data Validation

- Designed generic SQL import scripts processing **1.6M HUD and Tax Credit income limit records** across 5 years; automated eligibility definitions based on family size, income levels, and AMI percentage thresholds; built data normalization and validation layers preventing data corruption in client environments.

56 State-Specific Housing Certification Documents

HUD COMPLIANCE PROPTECH

PHP · PostgreSQL · AWS S3 · Adobe Acrobat DC Pro · Merge Fields · Document Management

- Created and maintained **56 state-specific affordable housing certification documents** with merge fields for Section 8/LIHTC/HUD certifications; integrated document storage and retrieval via AWS S3; provided technical specifications and quality assurance testing for each state variant.

Leasel: AI Compliance System for Residential Lease & Affordable Housing

AI + COMPLIANCE

Python · LangGraph · ChromaDB · Deterministic Rules Engine · Compliance Automation — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **93.6% precision** on 1,240 residential lease PDFs across 10 compliance rules (5 US states + federal standards); deterministic rules engine ensures fully auditable compliance determinations; demonstrates ability to automate and validate complex housing compliance logic at scale.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87

Data engineering, statistical modeling, AI pipelines, data governance, anomaly detection

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

HUD & Compliance Domain: HUD Payment Processing, HAP Allocation, LIHTC / Section 42, HCV / Section 8, Income Eligibility Verification, AMI Threshold Calculations, Recertification Workflows, Housing Subsidy Compliance, Affordable Housing Regulations, Rent Schedules, Tax Credit Calculations, State-Specific Certification Documents

Database & Technical: SQL (expert — 200+ production optimizations, data fixes, data normalization, execution plan tuning, indexing, partitioning, triggers, audit tables, PL/SQL), PostgreSQL, MySQL; PHP, Python, JavaScript, TypeScript, HTML/CSS; Webservices, XML, REST APIs, AWS (S3, EC2, RDS), Docker, CI/CD (GitHub Actions), Git, Linux

Support & Operations: Client Application Troubleshooting, Issue Escalation & Resolution, Quality Assurance Testing, Technical Documentation, Module Enhancement & Bug Fix Documentation, Internal Debugging Tools, Data Corruption Analysis & Data Fixes, 300+ Code Reviews, 5+ Engineers Mentored, Agile/Scrum